



## AQ8.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Consider two vectors  $a = (2, 0, 3, 0, 8)$ ,  $b = (3, 2, -2, 4, 0)$  in  $R^5$ .

Choose the set of correct options.

☐

$a$  and  $b$  are orthogonal.

☐

$a$  and  $b$  are not orthogonal.

☐

$(a - b) \cdot a = 0$ .

☐

$(a - b) \cdot a = 77$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$a$  and  $b$  are orthogonal.

$(a - b) \cdot a = 77$ .

1 point

Choose the set of correct statements.

☐

In an orthogonal set, the norms of all the vectors are equal.

☐

In an orthogonal set, the vectors are linearly independent.

☐

In an orthogonal set, the vectors are linearly dependent.

☐

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $R^n$ , then there must be a unique solution to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

☐

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $R^n$ , then there are no solutions to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

☐

The determinant of a square matrix formed by a set of orthogonal vectors in  $R^n$  is zero.

☐

A set of  $n$  vectors can never form an orthogonal basis in  $R^{n-1}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

In an orthogonal set, the vectors are linearly independent.

If the columns of an  $n \times n$  coefficient matrix  $A$  comprises of the individual vectors of an orthogonal set in  $R^n$ , then there must be a unique solution to the system  $AX = b$ , where  $X, b$  are  $n \times 1$  vectors.

A set of  $n$  vectors can never form an orthogonal basis in  $R^{n-1}$ .

1 point

Which of the following is an orthogonal basis of the given vector spaces with respect to the standard inner product (dot product)?

☐


$\{(1, 0), (0, 1)\}$  is an orthogonal basis of  $R^2$ .

☐

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$  is an orthogonal basis of  $R^3$ .

☐

$\{(3, 4), (4, -3), (2, -3)\}$  is an orthogonal basis of  $R^2$ .

☐

$\{(2, 1, -1), (-1, 1, -1), (3, -3, 3)\}$  is an orthogonal basis of  $R^3$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(1, 0), (0, 1)\}$  is an orthogonal basis of  $R^2$ .

$\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$  is an orthogonal basis of  $R^3$ .

## Level 2

1 point

Find a vector in  $R^4$  that is orthogonal to the subspace spanned by  $(1, 1, 0, 0)$  and  $(0, 1, 1, 0)$  with respect to the dot product as the inner product.

☐  $(1, -1, 1, 0)$ 
☐  $(2, 3, 4, 5)$ 
☐  $(1, -1, 1, 1)$ 
☐  $(1, 1, 1, 0)$ 

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(1, -1, 1, 0)$

$(1, -1, 1, 1)$

1 point

Which of the following are orthogonal to the vector  $(1, 2)$  in  $R^2$  with respect to the inner product  $\langle v, w \rangle = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$ ?

☐

$(-7, 1)$

☐

$(6, -1)$

☐

$(7, 1)$

☐

$(-9, 1)$

☐

$(14, 2)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(7, 1)$

$(14, 2)$

Consider the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5 \quad x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9 \quad 3x_1 - 6x_2 - 5x_3 = 9.$$

Number of solutions of the above system is

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Let  $AA$  be the coefficient matrix of the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1 \quad x_1 - 2x_2 + 3x_3 = 1$$

$$2x_1 + x_2 = 5 \quad x_1 + x_2 = 5$$

$$3x_1 - 6x_2 - 5x_3 = 9 \quad 3x_1 - 6x_2 - 5x_3 = 9.$$

Which one of the following is true about the matrix  $AA^T$ ?

- ☐ A scalar matrix.
- ☐ The identity matrix.
- ☐ A diagonal matrix.
- ☐ A lower triangular matrix.
- ☐ An upper triangular matrix.
- ☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

A diagonal matrix.

A lower triangular matrix.

An upper triangular matrix.

[Check Answers](#)

Your score is: 0/7